

Experiment 8 — Data Lifecycle Management Policies for Storage Optimization and Energy Efficiency

Step 1: Create an S3 Bucket

1. Log in to **AWS Management Console**
2. In the search bar, type **S3** and click on it
3. Click **Create Bucket**
4. Enter a globally unique bucket name (e.g., **d1m-experiment-bucket**)
5. Region: **Asia Pacific (Mumbai) — ap-south-1**
6. Under **Object Ownership** — keep **ACLs disabled**
7. Under **Block Public Access** — keep **Block all public access** checked
8. Under **Bucket Versioning** — leave disabled
9. Under **Default Encryption** — leave as **SSE-S3** (enabled by default)
10. Click **Create Bucket**

Screenshots needed:

- Bucket creation configuration screen
 - Success confirmation showing your bucket in the bucket list
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Step 2: Upload Sample Data

1. Click on your newly created bucket to open it
2. Click **Upload** → **Add Files**
3. Upload the following three files:
 - A **PDF document** (e.g., any experiment file) — represents frequently accessed data
 - A **text file** (.txt) named something like **backup.txt** — represents rarely accessed data
 - An **Excel or any other file** (.xlsx) named something like **myExcel.xlsx** — represents archive data
4. Click **Upload** and wait for success confirmation

 **Screenshot needed:** Upload status screen showing all files as "Succeeded", and the bucket objects view showing all files listed with their types and sizes

Step 3: Data Classification Table

Note the following classification table in your journal based on your uploaded files:

File Name	Type	Access Frequency	Category
[Your PDF name]	PDF	Frequent	Hot Data
backup.txt	Text	Rare	Cold Data
myExcel.xlsx	Excel	Rare	Archive Data

Step 4: Configure Lifecycle Policy

1. Click on your bucket name to open it
2. Click on the **Management** tab
3. Under **Lifecycle Rules**, click **Create Lifecycle Rule**
4. Enter a **Lifecycle Rule Name** (e.g., dlm-rule)
5. Under **Rule Scope** — select **Apply to all objects in the bucket**
6. Check the acknowledgement checkbox that appears
7. Under **Lifecycle Rule Actions**, check the following:
 - ☒ **Transition current versions of objects between storage classes**
 - ☒ **Expire current versions of objects**
8. Under **Transition current versions of objects between storage classes**:
 - Click **Add Transition**
 - First transition:
 - Storage class: **Standard-IA**
 - Days after object creation: **30**
 - Click **Add Transition** again
 - Second transition:
 - Storage class: **Glacier Instant Retrieval**
 - Days after object creation: **90**
9. Under **Expire current versions of objects**:
 - Days after object creation: **365**
10. Click **Create Rule**



Screenshots needed:

- Lifecycle rule creation screen showing transitions and expiration configured
 - Lifecycle Rules list showing your rule as **Enabled** with scope "Entire bucket"
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Step 5: Verify Lifecycle Rule

1. Click on your newly created rule name (e.g., dlm-rule) to open its details

2. Observe the **Review transition and expiration actions** section showing:

- Day 0 → Objects uploaded
- Day 30 → Objects move to Standard-IA
- Day 90 → Objects move to Glacier Instant Retrieval
- Day 365 → Objects expire



Screenshot needed: Lifecycle rule detail page showing the full transition and expiration timeline

Final Summary:

- Create S3 bucket and upload objects
- Go to Management and create lifecycle rule
- Transition Standard = 30 , Glacier = 60
- Expiry = 90 and create rule